

Agentic Robot Control:

How Should an Agent Participate in Robot Control?

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摘要

General-purpose agents are becoming increasingly capable of reasoning about visual environments, remembering previous interactions, and revising plans after observing outcomes. In parallel, robot foundation models and vision-language-action (VLA) policies are improving the generation of executable motions, while real-world reinforcement learning (RL) provides a way to adapt those policies through physical interaction. The central challenge is how to connect these capabilities in a reliable robot control system. This proposal studies *agentic robot control*: the allocation of control authority between an open-world reasoning agent, a robot policy, and a low-level controller. The proposed research starts from a conservative but broadly useful setting in which the agent observes complete execution trajectories after an episode, identifies task stages, progress, and failure onset, and provides structured supervision to the policy critic. The project then investigates how the same execution understanding can support policy orchestration, retry and recovery, and eventually the generation of spatial action targets. Experiments will use a Unitree G1 humanoid with two Revo2 dexterous hands and a frozen GR00T base policy, with real-world RL used to test whether agent-derived execution understanding improves policy adaptation under a fixed interaction budget.

1 Research Vision: From Robot Policies to Agentic Robot Control

Robotics is moving from isolated task policies toward systems that must operate in open-ended, long-horizon environments. A policy can map observations to actions, but reliable physical behavior also requires the system to remember what has happened, understand which stage of a task it is in, distinguish progress from apparent motion, and decide how to respond when execution goes wrong. These requirements become particularly important for humanoid manipulation, where contact, balance, body motion, and dexterous hands are tightly coupled.

The key question is therefore not simply whether a language model can control a robot. It is how an agent should participate in a control system whose components operate at different time scales and have different strengths:

How should an agent participate in robot control, and through which interfaces can its understanding of the world improve physical execution?

The proposal is based on a division of control authority. At the task time scale, an agent interprets instructions, decomposes goals, and maintains an interaction history. At the execution time scale, it may select a policy, compose policies, or produce a spatial representation such as a waypoint, visual prompt, or action chunk. At the learning time scale, it can inspect a complete trajectory and explain what happened, enabling a policy to learn from real interaction. The agent is therefore not introduced as a replacement for the low-level controller. Instead, its authority is increased gradually through interfaces that preserve the policy’s advantage in high-frequency physical execution:

understand execution $\rightarrow$ supervise policy learning $\rightarrow$ orchestrate policies
 $\rightarrow$ generate executable targets.

This staged view makes the research plan both ambitious and testable. The first stage asks whether an agent can understand physical execution well enough to improve a critic. Later stages ask whether that understanding can be used at run time to choose, retry, and combine policies, and whether it can eventually be converted into direct action targets. A negative result at an earlier stage is still informative: it identifies the interface at which agent reasoning is not yet reliable enough for the next level of control authority.

2 Related Work

The related work is organized as a progression of capabilities rather than as a list of independent models. Each work addresses one part of the path from world understanding to physical execution.

2.1 VISTA: Persistent Understanding of an Interactive World

VISTA studies how a general multimodal model can reason over an unfamiliar interactive environment while retaining a lossless visual memory of previous observations. The system combines raw visual input, free-form reasoning, active visual inspection, and persistent interaction history. Its evaluation is conducted on interactive visual games rather than robots, but the underlying design addresses a central requirement for agentic control: a decision should depend on what happened earlier, not only on the current frame. For humanoid manipulation, this suggests representing an episode as a sequence of actions, observations, and state changes that an agent can query when locating progress and failure.

Source. Qiushi Han, Keya Hu, Linlu Qiu, Cathy Wu, and Kaiming He, “VISTA: A Visual Harness for Reasoning in an Interactive World,” MIT project page, 2026; formal peer-reviewed

venue not verified. <https://vista-research.github.io/>

2.2 Claude Plays Robotics: The Control Interface Shapes Capability

Claude Plays Robotics compares several ways of connecting a general-purpose model to robots, ranging from low-level torque control and programmatic controllers to reinforcement learning and high-level commands to pretrained robot policies. The study shows that the apparent robotics capability of an agent depends strongly on this interface. High-level policy interfaces are more practical for difficult embodiments, while direct control remains challenging for high-degree-of-freedom robots. This motivates our central design question: which decisions should be made by the agent, and which should remain inside a robot policy and its low-level controller?

Source. Shmuel Berman, Michael Ilie, Jia Deng, and Daniel Freeman, “Claude Plays Robotics,” Anthropic Research Report, 2026; non-peer-reviewed report. <https://www.anthropic.com/research/claude-plays-robotics>

2.3 Harness VLA: Orchestrating Frozen Policies

Harness VLA treats a frozen VLA as a retryable local primitive and assigns higher-level reasoning to an agentic harness. The harness can handle semantic re-grounding, staging, task organization, and recovery, while the VLA remains responsible for contact-rich manipulation. This is an important intermediate point between a monolithic VLA and direct agent control: the agent expands the operating range of a pretrained policy without replacing its local visuomotor skill. The remaining question is how to learn from the execution history that the harness observes, rather than using that history only for run-time retries.

Source. Yixian Zhang et al., “Harness VLA: Steering Frozen VLAs into Reliable Manipulation Primitives via Memory-Guided Agents,” arXiv:2607.08448, 2026; arXiv preprint. <https://arxiv.org/abs/2607.08448>

2.4 VIA: Agent-Generated Spatial Control Targets

VIA recasts robot manipulation as an agentic visual-interface task. A general-purpose vision agent observes a 3D interface, issues intuitive spatial commands, observes the result, and revises its plan. This direction demonstrates that an agent can participate in action selection without robot-specific fine-tuning, provided that the interface exposes an appropriate spatial abstraction. Its current setting is primarily quasi-static manipulation, so the challenge for future work is to extend the same idea to dynamic, contact-rich, and whole-body execution.

Source. Hengyuan Hu, Priya Sundaresan, Jensen Gao, and Dorsa Sadigh, “VIA: Visual Interface Agent for Robot Control,” arXiv:2607.11119, 2026; arXiv preprint. <https://arxiv.org/abs/2607.11119>

2.5 VP-VLA: Visual Prompts as an Agent–Policy Interface

VP-VLA investigates a two-system architecture in which a System 2 planner decomposes instructions and produces structured visual prompts, while a System 1 VLA executes the resulting guidance. The work is closely related to VIA but makes the interface more explicit: the agent does not necessarily issue the final action, but produces a spatial representation that the VLA can consume. This supports a broader design space for DexbotRL in which the agent may communicate through visual prompts, waypoints, or action chunks rather than through raw joint commands.

Source. Zixuan Wang et al., “VP-VLA: Visual Prompting as an Interface for Vision-Language-Action Models,” arXiv:2603.22003, 2026; arXiv preprint. <https://arxiv.org/abs/2603.22003>

2.6 GR00T: A Foundation Policy for Humanoid Execution

GR00T N1 provides an open foundation model for generalist humanoid robots. It combines visual-language understanding with an action-generation module and is trained from heterogeneous mixtures of robot trajectories, human videos, and synthetic data. In the proposed system, GR00T represents the execution side of the agent–policy division: it supplies a strong action prior and handles continuous motion, while the agent provides task-level reasoning and post-episode execution analysis. GR00T is therefore a foundation policy that an agent can call, supervise, or adapt, rather than a complete agentic control architecture.

Source. NVIDIA, Johan Bjorck et al., “GR00T N1: An Open Foundation Model for Generalist Humanoid Robots,” arXiv:2503.14734, 2025; NVIDIA technical report / arXiv preprint. <https://arxiv.org/abs/2503.14734>

2.7 RL Token: Efficient Adaptation through Real Interaction

RL Token studies how a pretrained VLA can be adapted with a compact reinforcement-learning interface. The method exposes an RL token and trains a small actor–critic head while anchoring the learned policy to the pretrained VLA. This shows a promising route for improving speed and task-specific precision with limited real-world practice. However, efficient policy updates do not by themselves explain why an episode failed. DexbotRL proposes to supply this missing execution understanding through structured agent supervision for the critic.

Source. Charles Xu, Jost Tobias Springenberg, Michael Equi et al., “RL Token: Bootstrapping Online RL with Vision-Language-Action Models,” arXiv:2604.23073, 2026; arXiv preprint. <https://arxiv.org/abs/2604.23073>

2.8 RoboReward: Vision-Language Rewards for Robotics

RoboReward develops a robotics reward dataset and vision-language reward models using real-robot corpora, counterfactual negatives, near-misses, and temporally clipped partial outcomes. Its direction is to replace brittle hand-designed rewards with a model that can judge progress from robot videos. This is a strong baseline for the proposed project. The distinction is that DexbotRL does not initially ask the agent to replace the environment reward with a scalar reward. Instead, the agent produces phase, progress, and failure-onset information that is consumed by auxiliary critic heads.

Source. Tony Lee, Andrew Wagenmaker, Karl Pertsch et al., “RoboReward: General-Purpose Vision-Language Reward Models for Robotics,” arXiv:2601.00675, 2026; arXiv preprint. <https://arxiv.org/abs/2601.00675>

2.9 DenseReward: Dense Progress from Synthesized Failures

DenseReward learns frame-level reward and progress signals by synthesizing physically realistic failures in simulation. The method addresses the sparsity of terminal success signals and can support model predictive control or reinforcement learning. It complements RoboReward by emphasizing dense temporal feedback and failure synthesis. For DexbotRL, it motivates a direct comparison between automatically learned scalar progress and an agent’s structured, evidence-linked interpretation of a complete episode.

Source. Yu Fang, Wanxi Dong, Jiaqi Liu et al., “DenseReward: Dense Reward Learning via Failure Synthesis for Robotic Manipulation,” arXiv:2607.13033, 2026; arXiv preprint. <https://arxiv.org/abs/2607.13033>

2.10 Synthesis of the Related Work

Taken together, these works cover persistent world understanding, control-interface design, policy orchestration, spatial action representations, humanoid foundation policies, real-world adaptation, and automated reward learning. They do not yet provide a single experimentally grounded path from an agent’s understanding of a complete physical trajectory to improved policy learning and, later, to runtime control authority. This proposal focuses on that bridge.

3 Core Research Questions

1. **Execution understanding:** Can an agent reliably identify task stage, progress, and failure onset from complete multimodal robot trajectories?
2. **Learning utility:** Can this structured understanding improve critic estimation and real-world policy adaptation under the same interaction budget, compared with TD-only and scalar reward baselines?
3. **Control authority:** Which interface—policy selection, retry and recovery, visual prompt, waypoint, or action chunk—allows an agent to improve execution while preserving the stability of the low-level controller?

The three questions form a staged program rather than three unrelated projects. The first establishes whether the agent understands execution; the second tests whether that understanding is useful for learning; and the third uses the resulting interface to study run-time control.

4 Proposed Research Plan

4.1 Stage 1: Agent Hindsight for Critic Supervision

The first stage separates agent reasoning from the high-frequency control loop. A frozen GR00T policy executes an episode on a Unitree G1 with two Revo2 dexterous hands. After termination, an asynchronous multimodal agent reads synchronized visual observations, robot proprioception, reference actions, executed actions, and available tactile evidence. It produces temporally localized stage, progress, failure-onset, and confidence records with references to the evidence used.

A deterministic compiler converts these records into auxiliary targets for the critic. The environment reward, Bellman target, and actor action contract remain unchanged. This isolation makes the first experiment interpretable: any improvement must come from the additional execution supervision rather than from changing the action space or allowing the agent to intervene during rollout.

4.2 Stage 2: Real-World Policy Adaptation

The second stage connects the structured supervision to real-world RL. The base policy remains frozen, while an RL Token or an equivalent compact representation supports a lightweight actor-critic update. The main comparisons will include TD-only learning, scalar VLM reward, dense progress reward, and the structured agent-supervised critic. Evaluation will use paired initial states, fixed training budgets, multiple seeds where feasible, and held-out task instances. The primary outcomes are task success, termination distribution, progress, and recovery behavior; training completion alone will not be treated as evidence of efficacy.

4.3 Stage 3: Runtime Agent–Policy Control

Once execution understanding and learning utility are established, the same task-state interface can be used at run time. The agent may select a policy, trigger a retry, compose several policies, or hand a spatial target to a VLA. Harness VLA, VIA, and VP-VLA provide three complementary interface templates: policy orchestration, direct spatial tools, and visual prompts to a policy. The project will compare these interfaces according to task horizon, contact richness, latency, and safety requirements rather than assuming that one interface is optimal for every task.

4.4 Long-Term Direction: Agent–Policy–World Closure

The long-term goal is a closed loop in which an agent maintains a task-level world model, a foundation policy provides executable motion, and real-world interaction updates both the policy and the agent’s understanding of the task. The proposal does not assume that the agent will replace the policy. Instead, it asks how much control authority can be transferred to the agent while retaining reliable physical execution.

5 Data, Evaluation, and Resources

The target platform is a Unitree G1 humanoid equipped with two Revo2 dexterous hands. The proposed full action representation is a 64-dimensional SONIC body component together with 6-dimensional left- and right-hand components, for a total of 76 dimensions. The software stack is based on a frozen GR00T base policy and RLinf-style real-world reinforcement learning.

Training data will combine existing robot demonstrations, simulated trajectories, and real interaction episodes. Data conversion and filtering will create versioned derived datasets; raw data will remain unchanged. Evaluation will distinguish technical qualification, development signals, and efficacy results. In particular, an agent-labeling pipeline or a successfully resumed training job will not be reported as a policy improvement without a matched evaluation denominator.

The project requires three resource classes: (i) a humanoid platform and dexterous hands for controlled real-world interaction, (ii) simulation and replay infrastructure for safe ablations and data analysis, and (iii) GPU capacity for frozen-policy inference, asynchronous agent analysis, and actor–critic training. Exact GPU counts and run budgets will be fixed after profiling the selected GR00T checkpoint, agent model, and rollout throughput; this proposal intentionally does not invent a hardware number before that measurement.

6 Expected Significance and Feasibility

The proposed direction is promising for three reasons. First, it builds on complementary progress rather than requiring a new foundation model from scratch: VISTA and related agent systems provide ideas for memory and execution reasoning, GR00T provides a humanoid action prior, and RL Token provides a compact route to real-world adaptation. Second, the staged plan has an early falsifiable milestone. If structured agent supervision cannot improve critic learning under a controlled interface, the project can stop or revise before attempting direct agent control. Third, a successful execution-understanding interface can support several future systems—policy orchestration, recovery, spatial prompting, and action-target generation—without committing prematurely to one control abstraction.

The expected contributions are therefore:

1. a formulation of agent participation in robot control as a staged allocation of control authority across task, execution, and learning time scales;
2. a reproducible agent-to-critic interface for converting complete physical trajectories into structured, temporally grounded supervision;
3. a controlled real-world evaluation of whether this supervision improves adaptation of a frozen humanoid foundation policy; and
4. a principled comparison of runtime interfaces through which an agent can orchestrate policies or generate executable spatial targets.

7 Timeline

Year 1: establish the trajectory schema, agent qualification protocol, and critic-supervision baseline in simulation and replay.

Year 2: run controlled real-world RL experiments and compare structured supervision with scalar reward and dense-progress baselines.

Year 3: study runtime policy orchestration, retry and recovery, and intermediate spatial interfaces such as visual prompts and waypoints.

Year 4: integrate the strongest interfaces into an Agent–Policy–World system and evaluate transfer across tasks and embodiments.

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